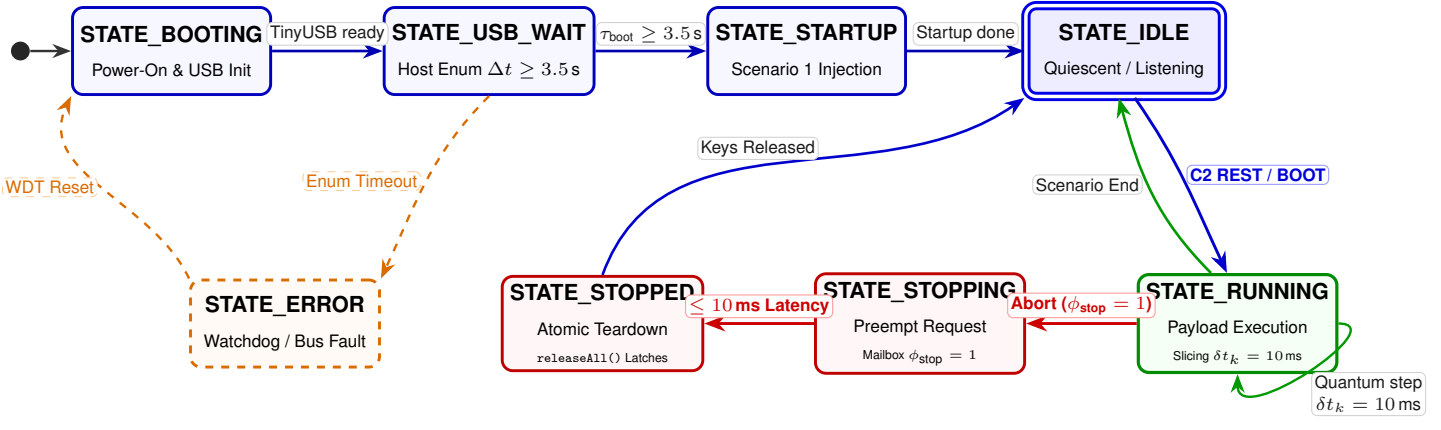

STATE_BOOTING
Power-On & USB Init
TinyUSB ready
STATE_USB_WAIT
Host Enum $\Delta t \geq 3.5\,\mathrm{s}$
$\tau_{\mathrm{boot}} \geq 3.5\,\mathrm{s}$
STATE_STARTUP
Scenario 1 Injection
Startup done
STATE_IDLE
Quiescent / Listening
WDT Reset
Enum Timeout
STATE_ERROR
Watchdog / Bus Fault
Keys Released
Scenario End
C2 REST / BOOT
STATE_STOPPED
Atomic Teardown
releaseAll() Latches
$\leq 10\,\mathrm{ms}$ Latency
STATE_STOPPING
Preempt Request
Mailbox $\phi_{\mathrm{stop}} = 1$
Abort ($\phi_{\mathrm{stop}} = 1$)
STATE_RUNNING
Payload Execution
Slicing $\delta t_k = 10\,\mathrm{ms}$
Quantum step
$\delta t_k = 10\,\mathrm{ms}$